HARSH JALNEKAR

Backend Engineer | Java & Spring Boot

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SUMMARY

Backend Engineer with over 2 years of combined experience building REST APIs and backend systems for a multi-tenant SaaS platform using Java, Spring Boot, Spring Security, and PostgreSQL. Experienced in authentication & authorization (Keycloak, OAuth 2.0, JWT, RBAC) and production debugging. Collaborates within Agile teams through code reviews and troubleshooting.

CORE SKILLS

Languages: Java, Python, SQL
Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate ORM
Databases: PostgreSQL
Messaging: RabbitMQ
Authentication & Security: OAuth 2.0, JWT, Keycloak, RBAC, AES/RSA Hybrid Encryption
Infrastructure & Tools: Docker, Git, GitHub Actions, Maven, Linux, Nginx, JMeter, CI/CD
Architecture: Microservices Architecture, Multi-Tenant SaaS Architecture, REST API Design, Exception Handling

PROFESSIONAL EXPERIENCE

NextGen Digi Lab — Pune, India | *Backend Engineer | Dec* **2024 – Present**

- **Multi-Tenant SaaS Backend —** Delivered backend development on independently developed RESTful services in Java/Spring Boot, improving long-term maintainability by structuring clean service and repository layers.
- **Scalability Design —** Designed the platform's architecture to support scaling from a single-organization deployment to 100+ tenant organizations without a structural rebuild.
- **Production Debugging —** Resolved a production database connection failure ("Too Many Clients"), confirmed fixed by the absence of repeat incidents, by replacing manual Hibernate session/transaction handling with Spring's @Transactional.
- **Database Design —** Enforced secure multi-tenant data isolation at the database level by designing tenant-aware PostgreSQL schemas and SQL queries.
- **Authentication & Security —** Closed a key security gap by implementing authentication and authorization using Keycloak, OAuth 2.0, JWT, and RBAC, and securing sensitive data with hybrid AES/RSA encryption.
- **Asynchronous Messaging —** Cut email-processing blocking by migrating delivery from synchronous JavaMailSender to asynchronous RabbitMQ-based processing, improving throughput by approximately 70%.
- **Code Reviews & Delivery —** Supported reliable delivery by participating in code reviews and Agile sprints, applying problem-solving to debug and troubleshoot production issues using Git, GitHub Actions, Docker, and Nginx.
- **AI-Assisted Tooling (In Progress) —** Building an on-premises document summarization feature using Ollama, aimed at reducing manual document review effort once released.
- **Performance Testing —** Used JMeter to identify performance gaps and threading issues under load, isolating root causes before deployment.
- **Deployment Automation —** Contributed Python-based deployment scripts that pull code from a specified branch and automatically deploy it to the server as part of the team's CI/CD workflow.

PRADO Pvt Ltd — Pune, India | *Java Trainee | Jan* **2024 – Oct 2024**

- **Monitoring Systems —** Kept a real-time hardware monitoring backend stable by managing multithreaded synchronization logic for sensor arrays in core Java.
- **Network Workflows —** Secured inter-lab communication by configuring Samba and proxy server workflows to manage data exchange between laboratory environments.
- **Debugging & Troubleshooting —** Improved reliability of real-time data transmission by refactoring network socket and stream handling logic.

EDUCATION

B.Tech, Electronics & Telecommunication Engineering — Vishwakarma Institute of Technology, Pune, India | 2020 – 2024

CERTIFICATIONS

Google Cloud Gen AI Academy – APAC Edition | Vector Search with AlloyDB – Google Cloud | Engineer AI Agents with ADK – Google Developer Program

LEADERSHIP & INITIATIVE

Advisory Board Member, Antariksh Astronomy Club, Pune (2022–2024) — Grew club membership from 7-8 to 50+ members and helped establish an inter-college astronomy community.
PRADO Pvt Ltd — Represented the department in a government audit.